A Catalog of Ghosts: the 120 Pilot Candidate Counterexamples and Sturmian Realizability Data

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Abstract

This catalog accompanies the Shadow Frontier project (successor to *Finite Spectral Shadows for the Collatz Valuation Cocycle*). It lists the 120 candidate-counterexample valuation words of pilot run 001 (deterministic regeneration, seed 1): words of length $M = 240$ over the alphabet $\{1, 2, 4\}$, calibrated to mean $\log_2 3$ within $2/M$ and carrying a non-coboundary order-3 shadow bias, sampled from the law $p_1 = 0.567$, $p_2 = 0.357$, $p_4 = 0.076$. Every one of the 120 is a *mixed-adic ghost*: its 2-adic realizability class modulo 2^{B_M+1} has least positive representative of nearly full bit length, not stabilizing between prefix lengths $M/2$ and M — the signature that no positive integer realizes the word. We also list ten Sturmian ghosts of the critical slope $\alpha = 2 - \log_2 3$ (all Sturmian words are now *provably* non-realizable, by the repetition-rigidity theorem); their data is shown for comparison. The objects below are points of $\mathbb{Z}_2$; the Collatz conjecture asserts that classes like these never meet $\mathbb{Z}^+$.

1 Conventions

For a valuation word $w = (b_0, \dots, b_{M-1})$, $B_N = \sum_{n<N} b_n$; the set of odd x whose first N valuations equal the prefix is a single residue class modulo 2^{B_N+1}; "rep bits" is the bit length of its least positive representative r_N, "mod bits" is $B_N + 1$, and the fill ratio is their quotient. A positive realization would force r_N to stabilize at $\lceil \log_2 x_0 \rceil$ once $2^{B_N} > x_0$; a ghost rides the diagonal $r_N \sim 2^{B_N}$. The bias is $|D_3| = \left| \frac{1}{M} \sum_n \zeta_3^{-b_n} - \mathbb{E}_{\text{geom}}[\zeta_3^{-b}] \right|$, the order-3 non-coboundary two-point shadow.

2 Translating between a word and the integer it represents

Explanation

A word does not represent a single integer but a single *residue class*: the set of odd x whose first M valuations equal $w = (b_0, \dots, b_{M-1})$ is exactly one arithmetic progression

$$x \equiv r \pmod{2^{B_M+1}}, \qquad B_M = \textstyle\sum_{n<M} b_n,$$

and "the integer" tabulated in this catalog is its least positive representative r. Every member $r + k\,2^{B_M+1}$ ($k \geq 0$) produces the same first M letters; the word sees only $x \bmod 2^{B_M+1}$. The converse direction is trivial: iterate $x \mapsto (3x+1)/2^{v_2(3x+1)}$ and read the exponents.

Formula

Along the prefix the orbit point has the exact affine form

$$x_n = \frac{3^n x_0 + A_n}{2^{B_n}}, \qquad A_n = \sum_{j<n} 3^{n-1-j}\, 2^{B_j}, \qquad B_n = \sum_{j<n} b_j.$$

The letter condition $v_2(3x_n + 1) = b_n$ is the congruence

$$3^{n+1}x_0 + A_{n+1} \equiv 2^{B_{n+1}} \pmod{2^{B_{n+1}+1}} \iff x_0 \equiv 3^{-(n+1)}\big(2^{B_{n+1}} - A_{n+1}\big) \pmod{2^{B_{n+1}+1}},$$

(3 is a unit in $\mathbb{Z}_2$), so each letter appends b_n new binary digits of x_0: a Hensel-style lift. After M letters the class modulus is 2^{B_M+1} and r is the lift's value reduced to $[1, 2^{B_M+1})$.

Python

The lift as implemented in `search/frontier.py` (`realize`); r is built letter by letter:

```
def realize(bs):                          # word -> (r, B+1)
    r, Mb = 1, 1                          # class rep and its bit-modulus
    A = 0; B = 0; P3 = 1                  # running A_n, B_n, 3^n
    for b in bs:
        need = B + b; mod = 1 << (need + 1)
        c    = (3*A + (1 << B)) % mod          # A_{n+1} mod 2^{B+b+1}
        x0   = (pow(3*P3, -1, mod)
                * (((1 << need) - c) % mod)) % mod
        if need + 1 > Mb:
            r = x0; Mb = need + 1
        A = 3*A + (1 << B); P3 *= 3; B += b
    return r, Mb

def word_of(x, M):                        # integer -> word
    out = []
    for _ in range(M):
        t = 3*x + 1; b = (t & -t).bit_length() - 1
        out.append(b); x = t >> b
    return out
```

Commands

```
cd shadow_frontier
# word -> integer (paste the letters as one string)
python3 -c "import sys; sys.path.insert(0,'search'); \
  from frontier import realize; \
  w=[int(c) for c in '111411211121...']; \
  r,Mb=realize(w); print(r, r.bit_length(), Mb)"
# integer -> word, plus the four-panel candidate image
python3 search/candidate_image.py --word 1,1,1,4,1,1,2,... --out viz/cand.png
python3 search/candidate_image.py --seed 687871 --steps 55 --out viz/c687871.png
```

Worked example: catalog word #1

For the word #1 above ($M = 240$, mean 1.583, $|D_3| = 0.214$): $B_M = 380$, modulus 2^{381}, and the least positive representative is the 381-bit integer

4527633547648404711572525100895098690435481551399990761
1280444502204362196575406814871934889952859875157437507273
11

whose first 240 Syracuse letters reproduce the word exactly (machine-checked). Its fill ratio is r's bit length over $B_M + 1$: $381/381 = 1.000$ — and the half-word prefix already needs 184 bits of 185: the representative *grows with the prefix* ($184 \to 381$ bits from $M = 120$ to 240) instead of stabilizing. That diagonal ride is the ghost signature of the Conventions section: the class contains no positive integer below 2^{380}. Contrast a true integer realization: the survival champion 687871 *is* the least representative of its own 50-letter class, and stays put as the prefix grows — stabilization versus diagonal is the whole test.

3 The 120 pilot candidates: statistics

All 120 are ghosts: no representative stabilized; fill ratios ≈ 1.

#	mean b	$\lvert D_3\rvert$	$P(16)$	B_{240}	rep bits	mod bits	fill
1	1.5833	0.214	225	380	381	381	1.000
2	1.5792	0.214	225	379	379	380	0.997
3	1.5792	0.214	225	379	378	380	0.995
4	1.5833	0.239	205	380	381	381	1.000
5	1.5917	0.220	213	382	383	383	1.000
6	1.5792	0.215	225	379	379	380	0.997
7	1.5917	0.215	225	382	383	383	1.000
8	1.5792	0.216	225	379	380	380	1.000
9	1.5792	0.229	214	379	380	380	1.000
10	1.5792	0.233	225	379	380	380	1.000
11	1.5792	0.229	225	379	379	380	0.997
12	1.5792	0.216	225	379	380	380	1.000
13	1.5917	0.233	193	382	382	383	0.997
14	1.5792	0.215	224	379	380	380	1.000
15	1.5917	0.216	208	382	383	383	1.000
16	1.5833	0.214	225	380	378	381	0.992
17	1.5917	0.215	225	382	380	383	0.992
18	1.5792	0.218	225	379	380	380	1.000
19	1.5917	0.220	225	382	383	383	1.000
20	1.5792	0.260	214	379	380	380	1.000
21	1.5792	0.272	208	379	380	380	1.000
22	1.5917	0.220	214	382	383	383	1.000
23	1.5875	0.228	225	381	382	382	1.000
24	1.5875	0.246	214	381	380	382	0.995
25	1.5917	0.222	211	382	383	383	1.000
26	1.5917	0.242	189	382	383	383	1.000
27	1.5792	0.222	222	379	376	380	0.989
28	1.5917	0.214	211	382	383	383	1.000
29	1.5875	0.215	225	381	382	382	1.000
30	1.5792	0.260	217	379	375	380	0.987
31	1.5917	0.214	216	382	382	383	0.997
32	1.5833	0.239	203	380	380	381	0.997
33	1.5792	0.218	225	379	377	380	0.992
34	1.5875	0.217	225	381	380	382	0.995
35	1.5917	0.214	224	382	379	383	0.990
36	1.5917	0.216	225	382	377	383	0.984
37	1.5792	0.220	225	379	380	380	1.000
38	1.5917	0.225	225	382	383	383	1.000
39	1.5875	0.221	225	381	381	382	0.997
40	1.5792	0.218	225	379	379	380	0.997
41	1.5917	0.225	206	382	380	383	0.992
42	1.5792	0.238	212	379	377	380	0.992
43	1.5792	0.215	225	379	375	380	0.987
44	1.5917	0.215	225	382	383	383	1.000
45	1.5875	0.217	225	381	379	382	0.992
46	1.5792	0.215	219	379	378	380	0.995
47	1.5792	0.229	223	379	380	380	1.000
48	1.5792	0.215	225	379	377	380	0.992
49	1.5917	0.215	225	382	383	383	1.000
50	1.5917	0.216	225	382	382	383	0.997

#	mean b	$\lvert D_3 \rvert$	$P(16)$	B_{240}	rep bits	mod bits	fill
51	1.5792	0.229	223	379	377	380	0.992
52	1.5792	0.248	206	379	380	380	1.000
53	1.5917	0.218	224	382	376	383	0.982
54	1.5792	0.215	225	379	380	380	1.000
55	1.5917	0.233	225	382	382	383	0.997
56	1.5917	0.279	194	382	383	383	1.000
57	1.5792	0.215	225	379	380	380	1.000
58	1.5792	0.218	225	379	377	380	0.992
59	1.5792	0.238	201	379	380	380	1.000
60	1.5792	0.214	225	379	380	380	1.000
61	1.5792	0.238	225	379	380	380	1.000
62	1.5833	0.230	213	380	381	381	1.000
63	1.5792	0.222	219	379	379	380	0.997
64	1.5792	0.215	219	379	379	380	0.997
65	1.5792	0.222	225	379	380	380	1.000
66	1.5917	0.216	196	382	383	383	1.000
67	1.5917	0.215	225	382	383	383	1.000
68	1.5792	0.216	225	379	372	380	0.979
69	1.5917	0.216	225	382	382	383	0.997
70	1.5917	0.216	225	382	383	383	1.000
71	1.5917	0.214	224	382	382	383	0.997
72	1.5792	0.214	225	379	380	380	1.000
73	1.5875	0.221	225	381	379	382	0.992
74	1.5917	0.215	222	382	383	383	1.000
75	1.5792	0.215	225	379	380	380	1.000
76	1.5792	0.225	225	379	380	380	1.000
77	1.5833	0.241	225	380	381	381	1.000
78	1.5917	0.215	225	382	382	383	0.997
79	1.5875	0.221	225	381	380	382	0.995
80	1.5875	0.217	225	381	381	382	0.997
81	1.5917	0.225	220	382	380	383	0.992
82	1.5833	0.214	222	380	381	381	1.000
83	1.5917	0.215	225	382	381	383	0.995
84	1.5917	0.218	225	382	383	383	1.000
85	1.5792	0.222	225	379	378	380	0.995
86	1.5792	0.238	225	379	380	380	1.000
87	1.5792	0.229	225	379	376	380	0.989
88	1.5875	0.227	225	381	382	382	1.000
89	1.5792	0.218	225	379	380	380	1.000
90	1.5917	0.216	225	382	383	383	1.000
91	1.5792	0.216	225	379	378	380	0.995
92	1.5833	0.225	225	380	381	381	1.000
93	1.5917	0.220	220	382	382	383	0.997
94	1.5792	0.214	225	379	380	380	1.000
95	1.5792	0.238	223	379	380	380	1.000
96	1.5792	0.238	225	379	379	380	0.997
97	1.5917	0.233	221	382	383	383	1.000
98	1.5792	0.229	219	379	380	380	1.000
99	1.5917	0.218	223	382	382	383	0.997
100	1.5792	0.215	225	379	379	380	0.997
101	1.5792	0.218	225	379	380	380	1.000
102	1.5792	0.218	216	379	376	380	0.989
103	1.5792	0.229	225	379	379	380	0.997
104	1.5875	0.217	225	381	382	382	1.000

#	mean b	$\lvert D_3 \rvert$	$P(16)$	B_{240}	rep bits	mod bits	fill
105	1.5917	0.229	211	382	383	383	1.000
106	1.5833	0.232	225	380	380	381	0.997
107	1.5792	0.218	225	379	379	380	0.997
108	1.5792	0.214	225	379	380	380	1.000
109	1.5917	0.220	225	382	381	383	0.995
110	1.5917	0.220	225	382	383	383	1.000
111	1.5792	0.229	225	379	380	380	1.000
112	1.5833	0.218	225	380	379	381	0.995
113	1.5792	0.248	225	379	379	380	0.997
114	1.5792	0.229	223	379	380	380	1.000
115	1.5917	0.220	224	382	383	383	1.000
116	1.5917	0.233	200	382	382	383	0.997
117	1.5792	0.222	225	379	378	380	0.995
118	1.5875	0.214	225	381	382	382	1.000
119	1.5792	0.218	224	379	380	380	1.000
120	1.5792	0.214	225	379	380	380	1.000

4 Sturmian ghosts (critical slope, now a theorem)

Words $b_n = 1 + \mathbf{1}\{\{n\alpha + \theta\} \geq \alpha\}$, $\alpha = 2 - \log_2 3$. By the repetition-rigidity theorem these are non-realizable for *every* slope and intercept; the data below shows the same diagonal-riding signature as the candidates.

θ	$\lvert D_3 \rvert$	B_{240}	rep bits	mod bits	fill	rep bits at N=100/200
0.000	0.447	380	381	381	1.000	159/316
0.100	0.447	380	380	381	0.997	157/318
0.200	0.447	380	377	381	0.990	159/317
0.330	0.447	380	381	381	1.000	155/318
0.500	0.453	381	379	382	0.992	159/317
0.610	0.453	381	376	382	0.984	160/318
0.750	0.453	381	382	382	1.000	160/317
0.900	0.447	380	381	381	1.000	160/318
0.057	0.447	380	381	381	1.000	159/318
0.085	0.447	380	381	381	1.000	159/318

Example Sturmian words (first 120 letters):

$\theta = 0.00$:

```
122121221212212121221212212121221212212121221212212122121212
212122121212212122121212212122121212212122121221212122121221
```

$\theta = 0.10$:

```
122121212212122121221212122121221212122121221212122121221212
122121221212212121221212212121221212212121221212212122121212
```

$\theta = 0.20$:

```
121221212212121221212212122121212212122121212212122121212212
122121212212122121221212122121221212122121221212122121221212
```

$\theta = 0.33$:

```
121221212122121221212122121221212122121221212122121221212212
121221212212121221212212121221212212121221212212122121212212
```

5 The negative cycles, computed explicitly

The three known negative cycles anchor the ghost families: each periodic valuation word below has its unique 2-adic realization at a *negative* integer, so the least positive representatives of its realizability classes ride the ceiling $2^{B_N+1} - |x|$ — the periodic ghost lines of the trajectory zoo. Throughout, $S(x) = (3x+1)/2^{v_2(3x+1)}$ and a period-p word w with sum B satisfies the affine cycle equation

$$x = \frac{A_w}{2^B - 3^p}, \qquad A_w = \sum_{j=0}^{p-1} 3^{\,p-1-j}\, 2^{B_j}, \quad B_j = b_1 + \cdots + b_j.$$

Negative cycles are exactly those with $2^B < 3^p$ (denominator negative).

$x = -1$: word (1), period 1

$$3(-1) + 1 = -2 = -2^1 \cdot 1, \qquad v_2 = 1, \qquad S(-1) = \tfrac{-2}{2} = -1.$$

Cycle formula: $p = 1$, $B = 1$, $A_w = 3^0 2^0 = 1$, and $x = 1/(2^1 - 3^1) = 1/(-1) = -1$. This is the 2-adic limit of the all-ones word: $b \equiv 1$ forces $x \equiv -1 \pmod{2^k}$ for every k.

$x = -5$: word $(1, 2)$, period 2

$$\begin{aligned} &3(-5) + 1 = -14 = -2^1 \cdot 7, && v_2 = 1, && S(-5) = -7,\\ &3(-7) + 1 = -20 = -2^2 \cdot 5, && v_2 = 2, && S(-7) = -5. \end{aligned}$$

Cycle formula: $p = 2$, $B = 1 + 2 = 3$, partial sums $B_0 = 0$, $B_1 = 1$:

$$A_w = 3^1 2^0 + 3^0 2^1 = 3 + 2 = 5, \qquad x = \frac{5}{2^3 - 3^2} = \frac{5}{8 - 9} = -5.$$

Letter mean $\frac{3}{2} < \log_2 3$: in the positive world this word ascends, the gentle green dotted line of the trajectory zoo.

$x = -17$: word $(1, 1, 1, 2, 1, 1, 4)$, period 7

$$\begin{aligned} &3(-17) + 1 = -50 = -2^1 \cdot 25, && S\colon -17 \mapsto -25,\\ &3(-25) + 1 = -74 = -2^1 \cdot 37, && -25 \mapsto -37,\\ &3(-37) + 1 = -110 = -2^1 \cdot 55, && -37 \mapsto -55,\\ &3(-55) + 1 = -164 = -2^2 \cdot 41, && -55 \mapsto -41,\\ &3(-41) + 1 = -122 = -2^1 \cdot 61, && -41 \mapsto -61,\\ &3(-61) + 1 = -182 = -2^1 \cdot 91, && -61 \mapsto -91,\\ &3(-91) + 1 = -272 = -2^4 \cdot 17, && -91 \mapsto -17. \end{aligned}$$

Cycle formula: $p = 7$, $B = 1{+}1{+}1{+}2{+}1{+}1{+}4 = 11$, so $2^B = 2048 < 2187 = 3^p$; partial sums $(B_0, \ldots, B_6) = (0, 1, 2, 3, 5, 6, 7)$:

$$A_w = 3^6{\cdot}1 + 3^5{\cdot}2 + 3^4{\cdot}4 + 3^3{\cdot}8 + 3^2{\cdot}32 + 3{\cdot}64 + 128 = 729 + 486 + 324 + 216 + 288 + 192 + 128 = 2363,$$

$$x = \frac{2363}{2^{11} - 3^7} = \frac{2363}{2048 - 2187} = \frac{2363}{-139} = -17 \qquad (139 \cdot 17 = 2363).$$

Letter mean $\frac{11}{7} \approx 1.571 < \log_2 3 \approx 1.585$: again sub-critical, again a positive-world ascender. Note how close $11/7$ is to $\log_2 3$ — the -17 cycle is a near-balanced word, the best rational approximant realizable at small height, and exactly the shape that the Baker cycle bound ($x_{\min} \le Cp^{\tau+1}$) governs.

6 The June 2026 constraints, and the surviving candidate list

Since this catalog was first compiled, the program proved a stack of further theorems (companion notes: narrow-band exclusion; two clocks; universal ghost clock; CIC round chain; faithfulness). This section records what they jointly do to the counterexample profile, and then lists what now survives.

The new constraints, in catalog form

- *Exact narrow-band exclusion.* No acyclic orbit dwells more than $CY^{\theta(D)}$ consecutive steps in a band $[Y, 2^{2D}Y]$; the count of confined words is an exact integer (no Perron estimates), and the counting capacity is exhausted at $D^* = 1.246$ (band ratio 5.63). Bounded-discrepancy words ($\sup|S_j| < 1.246$) are unconditionally non-realizable by divergent orbits — the Sturmian section above is now a corollary. In a ratio-≤ 2 band the dwell bound is *logarithmic* (the dyadic strip carries one word per phase).
- *The phase diagram has two arithmetic lines.* Vertical: $D^* = 1.246$. Horizontal: the binary bias ceiling $\varepsilon_{\text{bin}} = 0.449$ — calibrated $\{1, 2\}$-words cannot carry more order-3 shadow bias than this; exceeding it requires letters $b \geq 3$, which are descent-heavy. Saturation $\varepsilon^* \approx 0.54$.
- *Every letter is a metered ghost approach.* b=1 runs meter -1 (length exactly $v_2(x+1)-1$); b=2 runs meter the trivial cycle $+1$ (depth decrement exactly 2, parity handoff); $b \geq 3$ letters read $v_2(x + \frac{1}{3})$. Universally: every signature σ of depth K has phantom root $\rho_\sigma = C_\sigma/(2^K - 3^\ell)$ (the cycles above are instances), alignment drops exactly K per ridden block, rides last exactly $\lfloor (a-1)/K \rfloor$ blocks, and alignment a at height x costs $2^a \leq |2^K - 3^\ell|x + C_\sigma$. Amortized over *all* ghost families of depth ≥ 3, the available drift gain is $R = 0.0882$ bits per letter against a 0.415-bit budget.
- *Supply is consumed without replacement.* An acyclic orbit visits each integer once: each dyadic band holds only 2^{k-g} points of depth g per ghost, ever. Consequences: at most 2^{k-3} upward crossings of level 2^k; a divergent orbit spends $O(X \log X)$ lifetime steps below X.
- *The survival coin.* Parsed at 1-run starts, life against the letters-≤ 2 regime is the reading $v_2(3^\ell m - 1) = v_2(m - 3^{-\ell})$, death iff odd. On Haar fibers the coin is exactly i.i.d. fair-thirds; on ghost-aligned seeds it is deterministic (LTE): Mersenne $2^g - 1$ survives round one iff g is odd with $v_2(g-1)$ even; the 3-smooth families $2^g 3^a \mp 1$ cascade in closed form, with round-two readings such as $v_2(a' + 2 - \text{Log}_3(-5))$ — the 2-adic *logarithm of the* -5 *ghost above* — and every tested seed dies by round 7.
- *The* $\{1, 2\}$ *alphabet has an entropy deficit.* $\theta_{12}(D) \leq 0.97907 < 1$ at every band width: for an orbit with no letter ≥ 3 the phase diagram's allowed region is empty — no band of any ratio offers refuge.
- *The blind spot bites.* A positive integer has $\log_2 x_0$ bits, period: after $\approx \frac{2}{3}\log_2 x_0$ steps the word has consumed them all and every later letter is forced by the $\times 3$ carry cascade. The infinite word has Kolmogorov complexity $\leq \log_2 x_0 + O(1)$ while repetition rigidity demands near-maximal factor complexity in banded windows. Realizable words are a countable, computable family (of the 2^{30} depth-30 words over $\{1, 2\}$, exactly 66 are realized below 10^6).

The composite profile

A counterexample must therefore be: a divergent orbit, fleeing upward (escape $O(X \log X)$, single excursions below X sublinear), never pausing in any narrow band, calibrated to mean $\log_2 3$ yet carrying non-coboundary bias below the ceilings (0.449 binary, ≈ 0.54 saturated), financing every ascent through without-replacement ghost-depth pools and an amortized 0.088-bit census margin, beating an i.i.d. fair-thirds coin forever on its +1-handoff parities, with a word of near-maximal local diversity generated — after a free opening of $\frac{2}{3}\log_2 x_0$ moves — by a fixed finite carry transducer with no further choices.

The candidate list, June 2026

No specific integer candidate exists: every concrete object ever computed in this program is either a 2-adic ghost or dies with a finite certificate. What survives is six structural profiles. For each we list its best known approximant and the meter that taxes it.

profile	best known approximants	taxing mechanism / status
C1. Carry-cascade pseudo-random divergent (generic profile: calibrated, biased, high-complexity)	the 120 pilot words of this catalog (all ghosts); integer side: survival champions 27, 4591, 31911, 517191, 687871 (depths $17, 27, 33, 38, 50$; 59–76% of each life is in the forced carry regime)	fair-thirds survival coin; census margin 4.7×; all known members die; champion height grows $\approx 2^{D/2.2}$ with depth
C2. Binary violator: letters $\{1,2\}$ forever ($\mathcal{C}_\infty \cap \mathbb{N}$, the CIC class)	none positive; the 2-adic members are the Sturmian-type ghosts of §3 and their high-complexity cousins	entropy deficit $\theta_{12} < 0.9791$: no band of any width is a refuge; parity coin must read even forever
C3. Parity-channel rider (word biased toward $(1,1,2)^\infty$)	integers 2-adically near $-19/11$; least representatives ride the full diagonal (ghosts)	the $K{=}4$ clock of $-19/11$: alignment a costs height $11x + 19 \geq 2^a$, rides last $\lfloor (a-1)/4 \rfloor$ blocks
C4. Ghost-aligned seeds (3-smooth neighborhoods $2^g 3^a \mp 1$, incl. Mersenne)	round-one survivors $2^5{-}1, 2^{13}{-}1, 2^{17}{-}1, \ldots$ ($g - 1 \equiv 2^{2t} \bmod 2^{2t+1}$); family champion $2^{705} - 1$ ($n = 704$: six rounds, readings $8, 2, 2, 2, 2, 2, 3$)	deterministic LTE cascade; ghost-log readings; every tested seed ($n \leq 3000$) certified dead by round 7
C5. High cycle (near-balanced periodic word, B/p a convergent of $\log_2 3$: $19/12$, $65/41$, $84/53$, $485/306, \ldots$)	the -1, -5, -17 cycles above (negative side); no positive candidate below current k-cycle exclusions	cycle formula $x = A_w/(2^B - 3^p)$ + Baker pinning $x_{\min} \leq Cp^{\tau+1}$ + narrow-band period localization
C6. Exponent-level conspirator (round-chain exponents tracking 2-adic digits of ghost logarithms, e.g. $N_0 = \mathrm{Log}_3(-5) = 2556611305511861515 \bmod 2^{64}$)	none known; alignment at exponent level costs height at exponent level	the lifted clock: ghosts-of-ghosts; p-adic Baker bounds the readings effectively

The list should be read as a specification of where a counterexample would have to hide, not as evidence that one exists: profiles C2–C6 are each pinned by an exact meter with a positivity cap, and C1 — the generic impostor — is the quenched calibration wall itself, the single open core to which every known framework (ours, Tao's, the LLM-collaboration manuscript audited in the companion notes) independently reduces.

7 Interlude: could a counterexample carry a message?

A speculation, wild but within reason. The counterexample signature is near-critical drift + persistent non-coboundary bias + high local word complexity + mixed $(2,3)$-adic coherence. Suppose someone wanted to *hide a message* in such an object. Could word complexity communicate information?

The channel, and its exact capacity

Any finite message can be written into an orbit's opening. Each scripted letter costs $\approx \log_2 3 = 1.585$ bits of seed (the cylinder modulus of §2), so a seed of n bits buys a free composition of $\approx 0.63\,n$ letters — and then the blind spot takes over: the class pins the seed, and *every subsequent letter is forced by the* $\times 3$ *carry cascade*. The total information content of an entire infinite Collatz word is $K(\text{word}) \leq \log_2 x_0 + O(1)$: a counterexample can only ever say one thing — its own name. The survival champions live this: 59–76% of their lives are in the forced regime, improvisation on a theme they no longer control.

Worked example: the 56-bit integer

`60822829639107739`

has a Syracuse orbit whose first 40 valuations spell GHOST in ASCII ($b{=}1 \mapsto 0$, $b{=}2 \mapsto 1$), at 1.475 seed-bits per scripted letter.

The information paradox

To survive, the stream must *look like* a random source of about 1.4 bits per letter (the calibrated entropy ceiling, less the Legendre tax t^* once the bias exceeds the free basin $\varepsilon_{\mathrm{nat}} \to 0.1397$), with near-maximal factor diversity in every banded window at scale $p \approx \log_2 Y$, forever — repetition rigidity tolerates nothing less. Yet its true information rate is asymptotically *zero*: finite seed, infinite output. By the completeness theorems of the faithfulness note, the statistical tests available to the dynamics are exactly the finite-shadow algebra; so a counterexample is a pseudorandom generator engineered to *fail precisely one named test* (the chosen non-coboundary character, where it must show constant bias) *while passing all the others* — built not by a cryptographer but by carry propagation. The budgets are quantitative: bias up to 0.1397 is free; every structured ghost channel combined supplies at most $R = 0.0882$ bits per letter of the 0.415 needed ($\leq 21\%$ efficiency); the rest must come from the unstructured remainder where no finite certificate sees it.

How such a word would function

A self-feeding stream cipher. State: the binary tape of x_n. Output: the v_2-readings at the bottom. Update: multiply by 3, propagate carries upward, truncate what was read. The opening $0.63 \log_2 x_0$ moves are the composer's; afterwards the machine's "randomness" is the binary digit stream of 3-power multiples of its own past — it reads its exhaust as fuel. By the round chain, each life-or-death reading is $v_2(3^\ell m - 1)$, so the functional requirement is that *the powers of* 3*, in base* 2*, along this one orbit, are non-normal in a precisely coordinated pattern.* That is the entire miracle, localized — the regime where mathematics has only logarithmic digit bounds (Stewart), and exactly where the program's wall stands. A counterexample would be a finite phrase whose forced elaboration is an unending, never-self-quoting, undetectably biased improvisation: a message that continues writing itself after its author has run out of ink.

Play with it

Spell any ASCII message into an orbit and read it back (from the project root `shadow_frontier/`):

```
python3 - <<'EOF'
import sys; sys.path.insert(0,'search')
from frontier import realize

def spell(msg):                       # message -> performing integer
    bits = ''.join(f"{ord(c):08b}" for c in msg)
    word = [1 if c == '0' else 2 for c in bits]
    return realize(word)[0]

def read(x, nchars):                  # integer -> message
    out = []
    for _ in range(8 * nchars):
        t = 3*x + 1; b = (t & -t).bit_length() - 1
        out.append('0' if b == 1 else '1'); x = t >> b
    s = ''.join(out)
    return ''.join(chr(int(s[i:i+8], 2)) for i in range(0, len(s), 8))

r = spell("GHOST")
print(r)                  # 60822829639107739
print(read(r, 5))         # GHOST
EOF
```

```
# render the four-panel candidate image of your message-orbit:
python3 search/candidate_image.py --seed 60822829639107739 \
    --steps 40 --out viz/cand_ghost_msg.png
```

Things to try: longer messages (cost ≈ 11.8 seed bits per character — `spell` of a 100-character message returns a ≈ 1180-bit integer, instantly); watch the height panel during the scripted prefix (your bit pattern *is* the drift: letters 1 ascend, 2 descend, so the orbit's silhouette draws your message's bit-density); read past the end of your message to watch the carry cascade take over the composition — the letters after $8 \cdot |\text{msg}|$ are the automaton's, not yours, and (the $(3/4)^D$ law) a random message's orbit takes a letter $b \geq 3$ within a handful of steps of being left to itself; try to compose a message whose forced continuation survives long — that is exactly the champion-hunting problem, and the house always wins.

8 Collatz Elevator Pitch

One-liner. "We can prove the average number loses this game. We can't prove that this particular number does — and every number is a particular number."

Thirty seconds. "The rule is simple: take a number — if it's odd, triple it and add one; if it's even, halve it. The conjecture says everyone eventually crashes down to 1. Now, the ups and downs of any number look exactly like coin flips, and the coin is rigged downward — so on average, everything falls. That part is basically understood. The problem is that nothing here is actually a coin flip. Each number's entire future — every up, every down, forever — was fixed the moment you chose it, written into its digits. So to prove the conjecture, you'd have to rule out that somewhere out there is one number whose predetermined script just happens to dodge the crash forever. Probability can't do that: probability talks about most numbers, never about that one. It's the same wall as: we can prove almost every number has perfectly shuffled digits, but nobody can prove $\sqrt{2}$ does — even though $\sqrt{2}$ is sitting right there, fully determined."

The twist, if she wants one more floor. "Here's what makes it maddening — and what the last year of work actually showed. We've cornered the hypothetical escape artist. It can't follow any pattern: we proved that if its moves ever repeat, even once, in the wrong circumstances, that repetition snaps shut into a loop, and all loops are dead. So a survivor would have to move in a way that's indistinguishable from pure chance — never repeating, perfectly balanced on the knife edge between rising and falling. But it also can't actually be random, because random walkers provably crash. It would have to be a perfect impostor: a fully scripted number flawlessly impersonating chance, forever, while hiding a secret bias it needs to stay alive. We're fairly sure that impostor can't exist — being biased and looking unbiased pull against each other — but proving a specific, predetermined thing can't perfectly imitate randomness is something mathematics doesn't yet know how to do. That's the whole obstruction, in one sentence: we can beat every strategy, but we can't beat 'no strategy.'"

A note on your original phrasing: "deterministic numbers that behave totally randomly" invites the response "then they crash, like random things do —done!" The fix is to put the burden where it really sits: deterministic numbers that we can't prove aren't perfectly disguised as random. The casino version also lands well: "We've proven the house always wins on average. The conjecture asks us to prove no individual gambler beats the house forever — when every gambler's every move was decided before they walked in."

A The 120 candidate words

Alphabet $\{1, 2, 4\}$, length 240, row-wrapped at 60 letters.

#1 (mean 1.583, $|D_3|$=0.214):
```
111411211121121111121144211141122221111222121121222121122221
112221122411212122222121212111211111121221222111411112211121
222111111112422222211111121211411212111214221114142121111141
211211412222121111111122222121141212112214111211111114422241
```

#2 (mean 1.579, $|D_3|$=0.214):
```
222222222222222211111121121142111121142111122122222111424121
111112412111211422111141112111122112212211111112121111211
212222111212111114221424111211121411211144212111114121114221
211211111111122121111121211221121211141121141242222122222111
```

#3 (mean 1.579, $|D_3|$=0.214):
222112111112112421221112111211112121221112111121121221144412
422222211122121211111112111111121141212111222422214111212111
112411112112224211112111211122121122111124112111224111121221
22412211411211112212241211242112221211144112111211212121111122

#4 (mean 1.583, $|D_3|$=0.239):
1111111111111111111111111111111111111212221141142141221114212
111421141112214212421122114411224111211122121221122222221111
111112221221121111111111121421121124112212121142244122121211
121411412211121111211221111422424421212112121142112221121111

#5 (mean 1.592, $|D_3|$=0.220):
1111111111111111111111111111121121111412414211121212411112242
121121121121211114211141222111242122111211222111111111122111
221222112412112111422412112222244212121411111124111221221212
212211211411111122112142111122211122224114212211221214211224

#6 (mean 1.579, $|D_3|$=0.215):
2222222242222222421121111141242121141122122222111112112241221
111121121111111111111221111122222421114212214211111112422221
11111222111112111222214111122221121211242111111411121211222
111112111112111112211221214211141122121141214111121211212112

#7 (mean 1.592, $|D_3|$=0.215):
121112111211212121111224121112121411212114112411221242222211
221221111112211111221212224214111211122112111222111211112112
212422141111221412421111222214112142112111121212111112211111
111141212222242111111221211221111212421212114112211222141211

#8 (mean 1.579, $|D_3|$=0.216):
122111111111121211211111411212222111112111112111111211142211414
222221141211111421222112121111112114122121211122121211421121
412121124112211112422112221111111411421211122222411112111221
214122121114112111211221141224111112112112412121221211222221

#9 (mean 1.579, $|D_3|$=0.229):
22222222222222422222222222222222222222224222111222211221144111
121121122111221112211111121114212411112221112112111122241111
2412121211411111211121111121221111122111211114121212114211111
111122112211111121111114224212112112111122111111111112112421

#10 (mean 1.579, $|D_3|$=0.233):
111112111222121222121211114111414222441214211212124111122211
111111111121121421111411122221211224211422141122144142111121
142141111212211111212222411111111111111122422121212211111111
111221422111212111111111142211121111212111211211121114122221

#11 (mean 1.579, $|D_3|$=0.229):
2222224222222422222222112211221241221221112211142211112122112
1111121112112121111211112211211211221122111122241121211241141
121211114211121121122111412121211214222111112112411211222121
224221212212212112111421211121412211111112211111111212211111

#12 (mean 1.579, $|D_3|$=0.216):
422211212114121122244111122221112411221221112121112441124421
112222111112111211112212111221221221211111111121242212112122
114122214121111111241241111211411212111111141112221141211212
2111121111211111111222421121114211111212212121121111212211221

#13 (mean 1.592, $|D_3|$=0.233):
1111111111111111111111111111111111111111111111111222212111121
122124214112211111112211111222122141114421122211212442114111
121241421122224122111422121211222421122114421142422121221111
211124144212122112121212221112111241112111121121114212122141

#14 (mean 1.579, $|D_3|$=0.215):
24222222222422222222111111121122111111111111111111221121224211
211121212222112122121412112122112122121121212111121142111121
212111221114111121121244122211211122111122124124112112222212
211211111111211214114111112124111224141124122121211111211411

#15 (mean 1.592, $|D_3|$=0.216):
1111111111111111111111111111111111112441112211121111112112211221
121221121141241214211241221111111224121122222211111222211122
122212222142121212142222241121411121121112221411414111122111
211211221121242112114124211211224114212222211141122222111111

#16 (mean 1.583, $|D_3|$=0.214):
2122214111112111212111112122111212111121122414224112111221122
241122142121211111124121124441122122111111122212124222211214
211121421111112122111124211221211412111111221122111111121222
114121211411211111124122212122121111212121221212112111211122

#17 (mean 1.592, $|D_3|$=0.215):
111211212242121222211221222111112122421112112112222214111212
2111112421224121121121211122111411221411211242211122221211411

411111421211221112112211112211121111111221221121221121114112
212221111111111411141121121411111122211212421112112222424212222

#18 (mean 1.579, $|D_3|$=0.218):
2222222242422111241111221221211411221221121212211221142411112
111224211221112121114111212114111112221211212124121111121222
121141211112222211121111122114124141122111221111212212221111
2122142211211211112121211111122211142112111221111211111122111

#19 (mean 1.592, $|D_3|$=0.220):
1124111112111121211124112121112122112111114122111112111112221
121212111212212112121114111111112141122122442121122112222112
211414211141112112212244111111211211112112221412411141212222
1111211221244112121111221212411212411212142422121122111111111

#20 (mean 1.579, $|D_3|$=0.260):
22222242222222222222222222222222222212122111112211111221124112
2111121122112241211141111222112122121122211111114121222122122
4211142121242212121121122111111211221111122111221242211111211
1212111111212221211112111111111111141121112411212222122212111

#21 (mean 1.579, $|D_3|$=0.272):
2222222222222222222222222222222222212121111221221112112222212121
1111112212112111122211112111121111122211121221111411214141212
1112112111214222112112121211224212122111124142121111121121121
121121112111111122121211221121122212222124412212111111412122

#22 (mean 1.592, $|D_3|$=0.220):
1111111111111111111111111111121111224212421114212412211111122241
221114214212211122221221414421121112221121141222111211112212
1141111121114112121111211112244112121112211212224211121111114
1211111212112121242422121212412122111224111112121221222211212

#23 (mean 1.587, $|D_3|$=0.228):
142222112411411114111211112112112241442212211111121121211222412
222121111211221211111211122114112121214111111211111221122122
1242212121111114121421121122211112121111141211124212111112112
1224111111111111111211112222211112242441112411421411111122111

#24 (mean 1.587, $|D_3|$=0.246):
1111111111111111111111211421221111111111111111111122122211412
121212122414211411111112142122114111211242121222224141142121
211414122212111221211121112111241111111221142111211112122141
212111142121111214124412121221222112221411121421141114211112

#25 (mean 1.592, $|D_3|$=0.222):
1111111111111111111111111111112422122111122221122421111111411
1211122212211124211422112412121221141111211221241122212414211
221112412222221212211122224224211212124211214222221211111122
211111121211221122122121112242221211122221111111122122112211

#26 (mean 1.592, $|D_3|$=0.242):
111111111111111111111111111111111111111111111111111121111111
221222221111122142241112242111114111111221112144222421121124
21424412112112211214112121224141142122111141112221112214121 1
1121122224421122122124122111111141242221221211412422122112111

#27 (mean 1.579, $|D_3|$=0.222):
2242222222222222222222211112221111212111211112111111112111111
12221122124222212212111111111221411121222121111142111 2141422
41112112111111112121122241141122112221111111111221112121121 12
221424122222421221111111111411112142222111111211422212121121

#28 (mean 1.592, $|D_3|$=0.214):
1111111111111111111111111111121114411214111212111211111121122
4112141121112221221122122122222224221224121112122121124121 11
1121121214212112111411122111242112411224212222211221121221 21
1212111222111121121442412111224422211212111122121114122112 12

#29 (mean 1.587, $|D_3|$=0.215):
1121124142111111111212422111111111111211122211141121124421121 2
4121121212221121222212112212111122111142112212221412121211 24
1112111124221121112424121112111222111112222121112212111121112
112212121211111442114414211211212112114211221212121222111111

#30 (mean 1.579, $|D_3|$=0.260):
22222222222222224222222222222222222222222222141111211211111112211
22111111121111124111222121122414212212122114112121121212121122
1111111122221112111211111111111112111111111114422212121214141221
21221121121112211111111121212111211222241221211111222111222211

#31 (mean 1.592, $|D_3|$=0.214):
111111111111111111111111112112211221112211421111221222222221214
212222111112121211122212112212222211111112212214124121121111
14112111211114111421144222421122224121111141211121144112222 1
112221421211111121112211242111224111122122412211222121421111

#32 (mean 1.583, $|D_3|$=0.239):
111111111111111111111111111111111111111211112411221142121421 1
241122242112114144211221211112111221121211121222112212122112
112211112111114121142142122114224221211214111221212111412212
221111112141221112111411444111421221241122111121111222122411 1

#33 (mean 1.579, $|D_3|$=0.218):
421241222121121121112121121222111121241111111122212112112241
222111121111112141221124222122112111211411111222112211121111
111111212211111122111221122221244111141242121122112214221112
121221211222212111221112211114212222211112422211111111124412

#34 (mean 1.587, $|D_3|$=0.217):
114111211111111411112144142414111211211122111112221112211112 2
112111222222211221122121212211221122111111112111124211221211
112412122121222224122421211111111214111221242211211212211412
211221211212111211114111111111111114114214112142222111121421121

#35 (mean 1.592, $|D_3|$=0.214):
111111111111111111412221112111212142144212212124221212124121 4
112112122421122111121411241222212242111221121222224111121212 2
111111121211212141212212112412412222111111411221112122241221
222111122111111211121111211224121122211111212411112111121111 11

#36 (mean 1.592, $|D_3|$=0.216):
111111111111121212222121211111212122114221411212222212112121 1
112112112212111111211412114212111221121141111111411112112421 1
111142112211422122114111411224211112124141221422221124214121
212411111122112121111211411112122122122211122411212211212211

#37 (mean 1.579, $|D_3|$=0.220):
222222241111111112121124422121211111211111111114121212411224
124211111211122121212111422111111212111111111122112414112212
112211214211121224211122121111111111212412411111142411211111
222121112242112241221111222212411112121112241211221111211121

#38 (mean 1.592, $|D_3|$=0.225):
111111111111124211211212112221221211112411212111121112411112
121222122442211222114242211211111222121141111211221211122142
141214442111124124224121121211121421214112111221111111241121
221111141112111211121111211111112211221142221111111121214112

#39 (mean 1.587, $|D_3|$=0.221):
224111212221141111112241222111411211111111141121211124121121 1
111211122221214111142211411221122112212111212212111112111111 1
241411211121242121211142222111114121121121111212121121111111
144221211111212412212211122211121121114221212111122444211221 1

#40 (mean 1.579, $|D_3|$=0.218):
222222222222114112211111121411121111211111211121111222411221 11
112121411111224111211112222122112211122112112212111141214221 2
222122122222122211211112111411222112121111214111122112221141 1
121111211212411412212212212121211411221441211111111124211121 2

#41 (mean 1.592, $|D_3|$=0.225):
111111111111111111111111111111111111122112111111221412214122 11
211112221112141221122111122211111112211121121421241424241211
422121111141111114111124111212121222111222122214122221221212
242121224111421442212111112221112124111141212242111112412212

#42 (mean 1.579, $|D_3|$=0.238):
222222222222222222222222222222211122111411211111111411241211 11
121241112411122112121111111211121121111122211112112212112121
211114121111112211112212212121211121211121121211111112122111 1
212241112411122122212211112211124222114112121212242222212441

#43 (mean 1.579, $|D_3|$=0.215):
222422242221121224211214141211211124411212111112111124111222
111112221111222412211111111112121211421121111212112224214221
221111212111122122112121111211111212211112221221121111112221 4
212241211211111221212111121121142211111114211111212221412121

#44 (mean 1.592, $|D_3|$=0.215):
111111111111111121112212211221111142111221111224211121211411
422211112211111222442111112211112122412111212222141221142221 1
112214211221111211111221224112121122122112112212111211111212 14
112112111121411112222114122221121212221224411442221222112112

#45 (mean 1.587, $|D_3|$=0.217):
121121122421221114141111111111121212241122111111111112111121 11
122112222211211212224111241141222241242141211221121221421412
111111122212212111422111121221112112114121211112222222211412
111111112212121121111121214111241111141121222112142211111141

#46 (mean 1.579, $|D_3|$=0.215):
222242222222222222222222212111112111121121422111441122111111 2
124111111112121111121112222111112122121121111221111224122114

211121144411111421221111222211111211111212212112111222222111111
214112141241214112114211112121211211121142112111112111121212

#47 (mean 1.579, $|D_3|$=0.229):
224222222222222222222224222244222222212211112411111441142111111
1111212112111112414121111121221212111122211212112112122211121 1
1114111212111221111111221212222112211212112111111111121221 1111
121421121122212242111222212222121221111111111121112121121411

#48 (mean 1.579, $|D_3|$=0.215):
242222222421221112111111111212111244114111111112242212142111 2
211121222212221241112112212212212112121221212421111141111211
121414111111111112111411211212222142121111212211212222121111221
242421221211111111111122221111222112111114112111111211221211 2

#49 (mean 1.592, $|D_3|$=0.215):
111111111141121142222122114411211221421214242121121244211111
212222112112112122142122114112121112122121211222222121211121 1
221111121122111111121121241112111221111122121211121211111122
222211112114121222211442142122211111241111222241212121111111

#50 (mean 1.592, $|D_3|$=0.216):
122112121421111242121122112121211221212221421421111121111121
141221111141421111222122121412122111114412111111111111121221
211144121112111211212211221122221211412121111111121114111212
121411242124212212122122124112211114121112112122421111221211

#51 (mean 1.579, $|D_3|$=0.229):
222222222122124211122412221122112112111221122221412211111212
211124411211121421112111411112211112224421111111121122211121
212122121111122222211122121141221222112122111211211122112221
111112122221141114114112121111121411222121121221122111212112

#52 (mean 1.579, $|D_3|$=0.248):
222422222222222222222222222222222222222222212211121112211111212 1
211122211111112111241111122221121111111222111121121122111211
242111111122112122421222112442112414121111221112211121111111
411112212211121211111124112211211212114411211121121121221121

#53 (mean 1.592, $|D_3|$=0.218):
111111111111111111421211144221112111212112114212214111112111 1
412122121422222222121111212112122222214111122121211141122212
214411114241111121422111221212111212111212121241111222111212
111212212112221121224222121221141112111221221122212222111211

#54 (mean 1.579, $|D_3|$=0.215):
242222224422222224422222222222421111121222111111121111111111142
211211112122221112121111142112221141111121122122112111122111
211242111221211111121211142121111141212121121111211111221122
111144221221211142211112112111122211111221121114121111211421

#55 (mean 1.592, $|D_3|$=0.233):
111111111111111121144111141412142211212111221111212211121111 1
221121211111222121111112211211214211122412111122212211141112
211112142112121241112214224112411221241211112112112111212112
221141111242241212141211111411211424222121122114114111121121

#56 (mean 1.592, $|D_3|$=0.279):
111111111111111111111111111111111111111111111111141112421241 12
241122111111121121142111114121224221111122121244211112111141
442122221411121121122112142112422411114211214211212411112411
211111122212122122141211111221214412211212211441411111211142

#57 (mean 1.579, $|D_3|$=0.215):
222222222211214112211211221111111211141122212224211211121111
111211421111222221121241222122141121121241111112111211241222
214122112112141111111112142112421111122121411121222141211211
122112111221422121111142222124211221121121112111212111111111

#58 (mean 1.579, $|D_3|$=0.218):
222222222222211111212111444121114121122121211111211221124124 4
122111411112112111211211111112122212142122111122121114411112
221121112211124221121122412221211111111142121112421112412112
121121122121112111122121211222112211122121112121122112221111

#59 (mean 1.579, $|D_3|$=0.238):
222222222222222222222222222222222222222222224222221122111111 2211
112112122112112144411211211114121241112112212111211212121111
112421114211422211211122122111121124111212114121111111121212
111112122111111211111421111221112111211111114211211121121211

#60 (mean 1.579, $|D_3|$=0.214):
222422222222222221122111242111211112412122111421112112111141221
111112112121411221111111112241211114212141141111122111141112
111211212221141411211144122121211121211221112421222111111222 21
111242112112111112121111211412111111221111122211221222222121 1

#61 (mean 1.579, $|D_3|$=0.238):
2222222222222222422111111111211411112121122211142421211111111
1211221121242114142211112222211122111111112122111111112112221
2122214212111122122211221221111211111111112214221112211112222
2121222212112111211422122121441221112112111221114211212121112

#62 (mean 1.583, $|D_3|$=0.230):
1111111111111111111111111111241221111121142121122421414111212
1112212221221411111211114222114121141114112121111212112122241
1122114124111121112242114121211212122244122421111412121111411
1221221111122112411211211111221112221211212112122122221114122

#63 (mean 1.579, $|D_3|$=0.222):
2222222222222222222222114121411211222112211112111122411122111
1112211111212122141422111221114112112411121122111121421122111
2121142141111112121111211114111421422222111211111212212411111
2221224111121121122111111111122121112111122212211222121112211

#64 (mean 1.579, $|D_3|$=0.215):
2222222222222222222222111111411112141211411111212122412111112
2111211211111214211111141222122121111144111111114112111242112
2111114122112211411222111211214111421121121122111111221212211
2212121121121111211111114112422112211212211121422111212221211

#65 (mean 1.579, $|D_3|$=0.222):
2222122211212111111412112141111222121111112121121221221222111
1112424112421211122111212111111122111211212222221121111212211
1121121221122122122111112211212222141121111212114122121211211
1421214211141122122222221211124112112111114142211211111441221

#66 (mean 1.592, $|D_3|$=0.216):
1111111111111111111111111111111111111111111111121212112211111111
2211222124122421122111121112111112212224111422212121122224211
1241124112114411211211122242141211411112211222222121212121122
2222112121142412124424211211121214212111242112422212121122111

#67 (mean 1.592, $|D_3|$=0.215):
1111121121114111111222212121114114111211211111212124114122221
1222221112122141141122121222212112114112121124112111122112212
2141112141112212211121411121111212112112212241214111222111121
4121111442112121111212412211212122212111212122211222111211222

#68 (mean 1.579, $|D_3|$=0.216):
2222222222222422424241112121111212412211442121121141111111221
2121121211111112442121122122222112121112111212112111412111112
1112121114122121111412111211111112212211122111111112112411212
1221211111141111111441122111111112242121121241112221111112111

#69 (mean 1.592, $|D_3|$=0.216):
1111111212241111221211114111112212112122241122212222211421111
1111142212112422121411222242422112142421112211112212112221111
1411111121212122112211111212211112211411111124221211121111111
1212411222112141111114111224211211121114121211211211111421212

#70 (mean 1.592, $|D_3|$=0.216):
2111421211224122241112241121121121121121121121121111141211222222
1112211121214121221212111111224112121121212111112211121422424
1122211212112121211422112221112112114111422211122241111111111
1241221211141211111221212211211121442111111442241211111111112

#71 (mean 1.592, $|D_3|$=0.214):
1111111111111111122112141214211121221122111141221442211222222
1144144124242112221411121121212111111211142122211122212122111
1222242112212111212222124111114112112211121121111212111111222
2112122112142121211112111221121241211211122141212111211111112

#72 (mean 1.579, $|D_3|$=0.214):
2422422222122111114111211111241121121121212211221111221212121
2244111142211221112241111112121211111211212114212121141242111
2121112112221122214122411112111111212214221212112122111122121
1222122121212112111112111214112111121111224111111112114111114

#73 (mean 1.587, $|D_3|$=0.221):
1211411111224111211211242211212122114112221111211111124214211
1222211111111221211121214121114411122121111212111111221212121
1122111111124124212112221221111141211241111111112111211211241
2111222422141111224112112111212221412121222411112211111411241

#74 (mean 1.592, $|D_3|$=0.215):
1111111111111111111212211212142222211411111221212112121122111
1112212211412421122221111221221212112122241211222221142212212
2122212211121211111222421221111111121212121211111211424414142
2214112111241112114111121111111221111412242212222114112111124

#75 (mean 1.579, $|D_3|$=0.215):
2211241114212121212121112222111111124212112121211211141221211
2121114121141111212111121141121241211241111211212212212224412

2211112222111121221221111112121212222211411121114222221111211
1142112211211212122121112122124124112121111212211211111114121

#76 (mean 1.579, $|D_3|$=0.225):
2222242111221411122111112421111111211121121241111112141111112
4211111211111212121111122412111214212121112111111111122211122122
2111221212212111411211124221242111241122112111111111112112414
2111412121121111121111212221214142211112112211144122144211112

#77 (mean 1.583, $|D_3|$=0.241):
1111111122221112111112112111122212114141222421422222221242142
2211221141242212121221121111241211121212111211211111121222221
1211221121211422221114222211112411112112111221111111221221122
1212212211242121112121122122121112212211122121111221212121112

#78 (mean 1.592, $|D_3|$=0.215):
111111111112112111212221144221122221222222111112112212111412
1122211222121142414111122212111121114111212441111121122222111
221222121121211121122214412211111111141212222111112122221222
1111224121111221112111121121111422122214214111122111111441121 1

#79 (mean 1.587, $|D_3|$=0.221):
1412112211111114212212121112121122222111211111111114111111111124
1121141422212211111211221421222212111211221111111144141 24222
1111112411214111211122111111242114121111421112212111221121 12
1241121411122211122121221121111241141112121221221121111222242

#80 (mean 1.587, $|D_3|$=0.217):
1111121442111112211141212112111211222111122121211112122111111 21
2112211111214142122121114221422121112112111211221121211 22142
1211122122121222211412222121211121111112212411111142111 22242
12211224222111241114212121211211242421212111221112111111121112

#81 (mean 1.592, $|D_3|$=0.225):
111111111111111111111112414111122222112141212121111141211421142 1
1122121111111211211212412111112242411111112211421121111212141
2222112111211422111222411124222111111141221121221241222111 12
1111121211122212112241142111242212421222142111111121412211121

#82 (mean 1.583, $|D_3|$=0.214):
21211212424121111214112121121221121121211212422111111111 21214
2241112111211244122211221221112221121211221411422141124244 21
12421122111211111211222411112112121211122111211111211221212 1
11111111122122111214112211121111222111112211224121112212211 11

#83 (mean 1.592, $|D_3|$=0.215):
1122121111242121122112111112112111114112212121222111111111122
22111122411122221222122212111121222211211211142224214412212 12
24111222122121112141412244222111224111121121211221111121114 21
41111212112112114141111122121221411211212111111222111212 1111

#84 (mean 1.592, $|D_3|$=0.218):
1221111111211411211222121422211422121122211112121221212112 11
12121111211212242141111114112124221111212222111121114411 2242
1211111111112212112212222212211441221111112122212211114111112 14
21111212142211421112121112111211411211111222121122222222111 2221

#85 (mean 1.579, $|D_3|$=0.222):
2121224411222211422212212121121111121111211141411412111111 22
122121111121121221211211211212211211412221211211121122221211
22212421221121111211211241212141111221241111221122121111112212
44122122121211112212122411121212122121222111211412121211111 1

#86 (mean 1.579, $|D_3|$=0.238):
22222222224222242222222222222112211111212241212211121242112214
11211111111111121122142111111221112111111212112141222111111 42
21122111111221111211121211121412212121211211111221212212211 2112
2121222211114114242122112121121111122222121211211141111112112

#87 (mean 1.579, $|D_3|$=0.229):
4222222222222222242224222214222211122212121112112111122111412
111221112111111422111221111112112411211112212241111111121111
11212221111111111111114111122411121122121111111122112112 42222
2211222211111121112112414211212211221222422111121212122 11221

#88 (mean 1.587, $|D_3|$=0.227):
2421412121211212121412111411122212121111121221121212121 22222
1211122211112111111212412111211122122111111112111112112122121
12124212241221211121212221111112212122221121242121212121221122
1122242111122221212442221111121414121111112114221111111124122

#89 (mean 1.579, $|D_3|$=0.218):
4222222222114111211221112121211221111111212111111111122122121
12112221121212212221111241211211141121142111121121211112111 4
2412211111221114111112241411212111211111242221222222221212224 2
2112211112112112211111112221121111212241114214111241222211 12

#90 (mean 1.592, $|D_3|$=0.216):
111111112112121114212212121112122112211121211112142211142111
211122111124412111222214212111111112121121121142211112421112
111121212212222121224142112211121212121212121112111411421211
111121141112212121111222221221411411112244142112111241224111

#91 (mean 1.579, $|D_3|$=0.216):
422222422221214111212112121214111211114111212111211111211112
112112121221241121112141221212111111111121122241112111122212
124412214142114412121121111112412211121111122221221212111211
2212211111124122111111112111221121111122121141211114214121222

#92 (mean 1.583, $|D_3|$=0.225):
111111212222111221221141121122112242122221111124221211112112
124111222411211241121122112111111112122412242211111111112212
211121111221212111211211111221211121212224222222222211122121212
1121121122212412424211412111121111111221111121242111241211221

#93 (mean 1.592, $|D_3|$=0.220):
1111111111111111111111221242121111114141212221211112112212111
112214111121221111112221241211211112412112121212212122112122
442421122111421124112121111221121112112114111244114111124122
221112111211222122121211424212242121411212111411122112122111

#94 (mean 1.579, $|D_3|$=0.214):
44222222222222224224121111111111111211122111121124142112211211
112112122121224212122411211112211122124212112111411121111211
141242112111111212211112224114121111124122111111111111142111112
2221112211111121211121421112122212211112112111121412212112211

#95 (mean 1.579, $|D_3|$=0.238):
222222222222224222222224222222222222222222112111111111112212121
2411111121111112112122111241121121111121421211211212111211221
1112412222111114111221221111114222412112111121211211111111112
211112212222111114111221222111124222411111111121111222412211111

#96 (mean 1.579, $|D_3|$=0.238):
22222222422222222222222121122421111211111212112124121211122122
22112222212111121111412211111111214112211122112121221111221112
121114122114111122111211221111222112124141211212124221211141
1222121111111211241221111122221211211111112211211112421122121

#97 (mean 1.592, $|D_3|$=0.233):
111111111111111111111121141422211112112211122222111112111142412
211211421421114212121224111422124111212421114221122212211112
112211211111112112121222111211114121211222214121411111211221
121411111222141111111222112211111411211412112411212244111112411

#98 (mean 1.579, $|D_3|$=0.229):
2222222222222222222222211114121111112211122122211111112121221
121211142211211112211111412221221111221112111111211121211111122
111212241411111111141222222121111121111112121121122222114111124
124121222411211211111242241221211421121111221112241121111212212

#99 (mean 1.592, $|D_3|$=0.218):
1111111111111111111111411221141424224211111241241111221121112212
12211114141121111221111241221211411211111122112124422221224221
1222121222121211111112112222242141222112211122221122111212212
122111211111212222121222211122111112111222221411111121121212211

#100 (mean 1.579, $|D_3|$=0.215):
222222211114411211242211211141212211222121411121211111121221
214112212122212222122212222111212142112212211124411411211111111
1144111114112111411411111142112211221121121112121212121212211
21211211111111212122121111112221122212212211111211121121411211

#101 (mean 1.579, $|D_3|$=0.218):
222222222222211111212212111112112111142121121211111221141241211
12111411111211212211111111212124112222112221121211111211122111
1422221121121121114111211112121242122222111142422212211421221
222411121111141214121212111121111121111212121121121111424122111

#102 (mean 1.579, $|D_3|$=0.218):
22222222222222222222222222242222222222222121211224211222122412112
21212111121121211122122114112111114112421111111211111211114111
1211221111121111112111114211211142111111111111224111244421121
11221111111112111211214111112111124241122111111122112121121121

#103 (mean 1.579, $|D_3|$=0.229):
22222222222121111121112111122411222221121211122122111221111122
122111121242112112121211121241212121121222442121112121442211
2112121211111112422121121121211242211112211111222211112121112
121412111211212122112221112412221241211142111111121241121112

#104 (mean 1.587, $|D_3|$=0.217):
121142222121212122121114211122111122122112221412112121222221
11112141121112111122114111111222121111122111111211111221222222

211421211211111211211111214111421411212221112211112142124122l
211111214122212114121211222112111114221141122121221224211141

#105 (mean 1.592, $|D_3|$=0.229):
111111111111111111111111111111121212211111212211221121141122l
1121211111111112112211122212224211121421124212221121222422222
112421212111112122211224411122222421222112111222124221222241
222112241121121241112212121212421122211121121112121241122212

#106 (mean 1.583, $|D_3|$=0.232):
111111211422221221212111241111121122224114221122122111422111
211122221211122112221222112112111122221211121212121242221122
212122111112112211111221211112221114224211214221122112412142
122141114121211211222111111211111224211122121111122111221112

#107 (mean 1.579, $|D_3|$=0.218):
242222222222122111111111122214121222111111221122111111111111
121211111111421211221111114222241121211112111111111121212221
222121222111121112121142211112121211412142112111221212212211
1121111111441221221111422221214122221112422211412222211111424

#108 (mean 1.579, $|D_3|$=0.214):
2422222222222222121221211221112121122122112111112224214414122
111211111121111141121111222111111122142111122111111142211122
111111122222222212221141411114142111212111111112112121411212
111122141441111222111112121112111111242112222111111211211141

#109 (mean 1.592, $|D_3|$=0.220):
1111111111111112242124214214112211121421222121122111242111214
211121211111221411414212111212421111111111221221111242142121
112142212222111121111112412422111111112121121111212211211212
412111111221111211121111111121211142422211412242122221112122

#110 (mean 1.592, $|D_3|$=0.220):
1111111111122214211212221221211111141121421112412411211142121
211111211111111112411212121111122222212411211111422221221122
2422222111111211421211111111211121122442111121111112121124211
112212211112111111214111221421142122122124211421214112111214

#111 (mean 1.579, $|D_3|$=0.229):
222222212122121121111212112111211111112114212221221111141212
1241121212221211111122121221124111112221221112112111211222111
1224141111111112211244412122121211111211224212222112122112221
112121221112211212112121212411212112211121242111211221422421

#112 (mean 1.583, $|D_3|$=0.218):
111111111112121112114122122141221112221111112111411111122111
2411214212111121212124114221124221111211112222144141214222121
112121121222111222112111211211111111212211111121121111121121
2124421114111222121221122211222111124424221112111144221211111

#113 (mean 1.579, $|D_3|$=0.248):
2222422222222222222221121222111222211121111142211121212111241
111112212121212242111121122211122112112212121111211221111111
112222112211112112122211211111422114212122112141411112211112
212411211212111121122112112214122221121422212121422211111112

#114 (mean 1.579, $|D_3|$=0.229):
222222222222222222211111211211211112121212124111122411112222
112222212211111211111211212111411222114124111111222112411211
211214212111111111114412112221221111212114212411212111221222
1422121211222221212221121211212121114111211211211211112214211

#115 (mean 1.592, $|D_3|$=0.220):
111111111111111111121121121114222221221222224221221121111114111
222212121112111111111144111422121212141112422141242112112112
121211411121221222111111241214422111222122121114121121241211
122211441112112121112111211111121111421111222411422211211111

#116 (mean 1.592, $|D_3|$=0.233):
11111111111111111111111111111111111111111111112112112114212221244
112111422124111121241222121142222121242122112121222111222212
221414412114111111121212421141214214121242122111111211112222
112111111112114121121112411111224222141121222124124111121111

#117 (mean 1.579, $|D_3|$=0.222):
22222222112222211114111112221421122121242211221111111122241221
21121121121122211121111212111121111121112221224211111121211422
111111121111212222411111121121122111112111214211111221121222
412222112212141111121412222211111212114124211241222114112221

#118 (mean 1.587, $|D_3|$=0.214):
124122242222141112121111112121211212211112121112111121422111
211221111121111111221141111222111122221211112111211111242212111
1112211242111221111141121112122111121421122422111222221224144
1242212222111114212111121111124241121222221111112112122121142

#119 (mean 1.579, $|D_3|$=0.218):
222242222222222242222222222222222421121121121411111111221142
111212111211121212221221121112212111221112112111111121124111
111211112211114211111112111111241421224221411112222121111122
141121122111111112112121141211221111112112111111141422241211

#120 (mean 1.579, $|D_3|$=0.214):
222222422224222222122411212122111121111111112121121121212111
212211121121112112111111211412211221112112211122241411112111
121122121211122114121121211412211111411112212111111211422444
411211111121211222411412212412411121211212111111121121211221